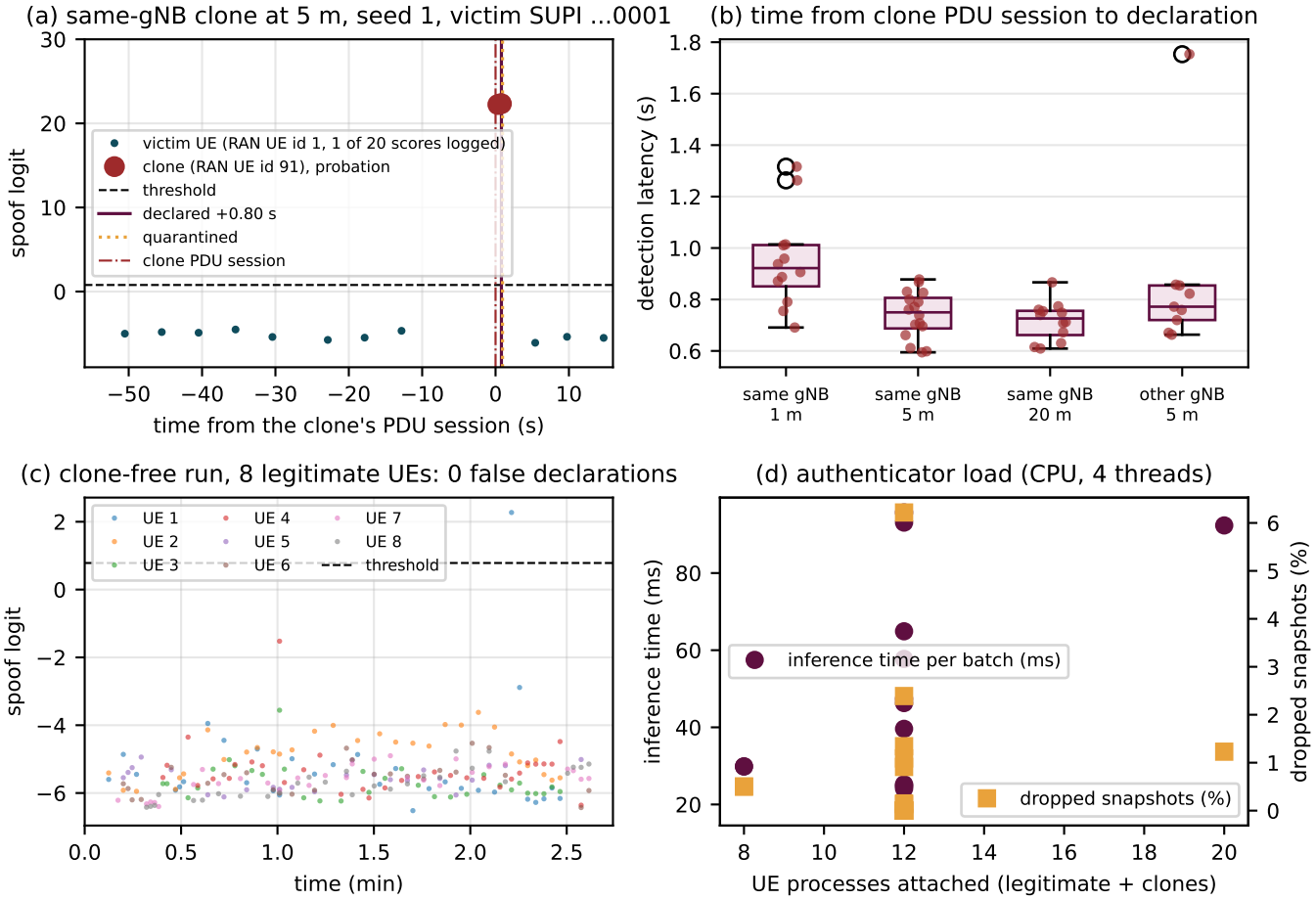
(a) same-gNB clone at 5 m, seed 1, victim SUPI ...0001
spoof logit
time from the clone's PDU session (s)
victim UE (RAN UE id 1, 1 of 20 scores logged)
clone (RAN UE id 91), probation
threshold
declared +0.80 s
quarantined
clone PDU session
(b) time from clone PDU session to declaration
detection latency (s)
same gNB 1 m
same gNB 5 m
same gNB 20 m
other gNB 5 m
(c) clone-free run, 8 legitimate UEs: 0 false declarations
spoof logit
time (min)
UE 1
UE 2
UE 3
UE 4
UE 5
UE 6
UE 7
UE 8
threshold
(d) authenticator load (CPU, 4 threads)
inference time (ms)
dropped snapshots (%)
UE processes attached (legitimate + clones)
inference time per batch (ms)
dropped snapshots (%)